

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Some stuff about me

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.

Education

2021.8-2025.6 **B.Sc. In Geographic Information Science**
Xi'an, Shaanxi

Shaanxi Normal University

Research Experience

2026.6-2026.11 **Research Assistant**
Hong Kong

The Hong Kong Polytechnic University

2024.8-2026.6 **Research Intern**
Guangzhou, Guangdong

The Hong Kong University of Science and Technology (Guangzhou)

Publications

1. Lyu, W., Dai, S., Song, Y., Zhao, W., Yi, W., Xiao, Y., & Jia, N. (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. <https://doi.org/10.1080/13658816.2026.2687121>
2. Lyu, W., Lei, Y., Yi, W., Song, Y., Li, X., Dai, S., Qin, Y., & Zhao, W. (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems*, 127, 102435. <https://doi.org/10.1016/j.compenvurbsys.2026.102435>
3. Lv, W., Lei, Y., Liu, F., Yan, J., Song, Y., & Zhao, W. (2025). gdverse: An R package for spatial stratified heterogeneity family. *Transactions in GIS*, 29(2), e70032. <https://doi.org/10.1111/tgis.70032>
4. Lv, W., Liu, F., Cai, K., Cao, Y., Deng, M., Liang, W., Yan, J., & Wang, G. (2024). Distinguishing the impacts and gradient effects of climate change and human activities on vegetation cover in the weihe river basin, china. *Journal of Geophysical Research: Biogeosciences*, 129(10), e2024JG008297. <https://doi.org/10.1029/2024jg008297>
5. Li, B., Lv, W., Ding, H., Song, Y., & Zhao, W. (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: Evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing*, 63(1), 2716413. <https://doi.org/10.1080/15481603.2026.2716413>
6. Xu, L., Shi, H., Lyu, W., Zhang, Y., & Zhao, H. (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography*, 136, 104789. <https://doi.org/10.1016/j.jtrangeo.2026.104789>
7. Chen, C., Song, Y., Lv, W., Shemery, A., Hampson, K., Yi, W., Zhong, Y., & Wu, P. (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. <https://doi.org/10.1111/mice.13489>

Note: Publications under “Lyu, W.” and “Lv, W.” both refer to Wenbo Lyu.

Honor

2024.12 **Longi Non-Education Major Scholarship**

2024.11 **First Prize in the 13th National University Student GIS Application Skills Competition**

2024.06	National University Student Innovation and Entrepreneurship Training Program Qualified Completion
2023.12	Grand Prize in the 12th National University Student GIS Application Skills Competition
2023.11	First Prize in the Second National University Student Ecological Environment Management Research Innovation Competition
2023.12	Second Prize of the 5th 'Guodi Cup' National College Student Natural Resource Science and Technology Competition, China Society of Natural Resources
2021.10	Outstanding Individual in Military Training Publicity for College Students, Shaanxi Normal University

Unpublished

Co-First Author (1st)	Quantifying pairwise directional influence between spatial variables: A reconstruction-based approach to asymmetric relationships	Submitted to AAAG, currently under second-round review
Co-First Author (2nd)	Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data	Submitted to IJGIS, currently under second-round review
First Author	Spatially convergent partial cross mapping for identifying direct causal influence in cross-sectional data	Submitted to Comms Phys, currently in revision
First Author	Unveiling interaction between spatial processes with information decomposition	In writing
First Author	Pattern Causality for Detecting Interactions with the R Package pc	In writing
First Author	coupling: Coupling Coordination Analysis in R	Plan
First Author	Measuring spatial stratified heterogeneity via information imbalance	Plan
First Author	Spatial causal discovery under latent confounders via deep orthogonal representation learning	Plan

Developed Spatial Analysis Toolkit

Package	Description	Language
spEDM	Spatial Empirical Dynamic Modeling	C++, R
tEDM	Temporal Empirical Dynamic Modeling	C++, R
infoxtr	Information-Theoretic Measures for Revealing Variable Interactions	C++, R
pc	Pattern Causality Analysis	C++, R
gdverse	Analysis of Spatial Stratified Heterogeneity	R, C++, Python
sdsfun	Spatial Data Science Complementary Features	R, C++
coupling	Coupling Coordination Analysis	C++, R